

On SOP Vibration Sensitivity in Hollow Core Fibers and the Impact of Multipath Interference

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Abstract A recent study reported two-order-of-magnitude higher SOP vibration sensitivity in hollow-core fibers (HCF). We demonstrate that multipath interference might greatly skew these measurements when using narrow-linewidth lasers. Utilizing a proposed robust characterization method, we show that HCF SOP sensitivity is actually comparable to solid-core fibers. ©2026 The Author(s)

Introduction

Vibration sensing is critical for fiber-optic links, enabling the identification and localization of high-risk events to facilitate proactive re-routing^[1]. While backward-sensing methods like Φ -OTDR offer high spatial resolution, their integration into active communication links is often hindered by the high cost of narrow linewidth lasers (NLL) and the need to mitigate power transients via complex continuous-probing^[2] or chirped-pulse^[3] architectures.

Forward-sensing methods^[4] offer a more compatible alternative for integrated sensing and communication (ISAC). Phase-based forward sensing, however, still requires NLLs to prevent phase noise from masking vibration signals. In contrast, state-of-polarization (SOP) sensing can be directly implemented by leveraging existing coherent receivers. This allows for seamless integration into active optical networks without the need for additional hardware or extra costs^[5]. Yet, in standard fibers, its sensitivity is orders of magnitude lower than phase-based methods, typically restricting its application to high-magnitude events like seismic activity^[6] or heavy train traffic^[7].

Nested anti-resonant nodeless fibers (NANFs) represent the state-of-the-art in hollow-core fiber (HCF) technology, offering record-low attenuation^[8], which has driven recent research interest in HCF-based sensing^{[9]–[11]}. Although theoretical models predict lower birefringence sensitivity in HCF compared to standard single-mode fibers (SMFs)^[12], recent field trials reported SOP sensitivity in NANFs two orders of magnitude higher than in SMFs^[11]. A confirmation of this finding would represent a breakthrough for long-haul sensing, paving the way for low-complexity and low-cost forward vibration sensing. However, the measurements undertaken in the study made use of NLL sources, which, in the presence of multipath interference (MPI) from SMF-HCF interfaces, may lead to parasitic phase-to-SOP con-

version, potentially overestimating sensitivity measurements. Regardless, a rigorous assessment of HCF polarization sensitivity is essential. A confirmation of higher sensitivity would enable straightforward vibration sensing in optical networks, while the alternative would benefit applications in polarization-insensitive interferometry^[13].

In this work, we conduct a thorough investigation on the impact of MPI in SOP vibration sensing in HCF, as well as provide new experimental results indicating similar SOP vibration sensitivity as SMF, contrary to the findings in^[11]. We provide theoretical and experimental evidence of MPI-induced limitations and evaluate the implications of employing narrow- versus broadband-sources.

MPI Impact on SOP Sensing

The polarization stability of a primary signal can be severely degraded by parasitic interference with a weak, time-delayed version of itself or an external crosstalk signal. This phenomenon, termed here as parasitic phase-to-SOP conversion, occurs when phase variations between two optical fields are mapped into fluctuations of the Stokes parameters.

Consider a primary optical field $\mathbf{E}_a$ and a significantly weaker parasitic field $\mathbf{E}_b$ (e.g., originating from multiple reflections). The total vector field $\mathbf{E}(t)$ resulting from their superposition is expressed in Jones notation as:

$$\mathbf{E}(t) = \mathbf{E}_a + \mathbf{E}_b e^{j\phi(t)} \quad (1)$$

where $\phi(t)$ represents the instantaneous phase difference between the two fields. This phase term encompasses environmental perturbations and mutual phase noise. The resulting SOP is characterized by the four Stokes parameters S_i ($i \in \{0, 1, 2, 3\}$). Substituting the field superposition into the definition of the Stokes vector yields:

$$S_i(t) = S_{ia} + S_{ib} + 2\text{Re}\{A_{i,int} e^{j\phi(t)}\} \quad (2)$$

Here, S_{ia} and S_{ib} represent the static Stokes parameters of the independent fields, while $A_{i,int}$ denotes the complex interference amplitudes. These amplitudes are determined by the cross terms of the Jones components (E_x, E_y) of the two fields:

$$A_{0,int} = E_{ax}^* E_{bx} + E_{ay}^* E_{by} \quad (3)$$

$$A_{1,int} = E_{ax}^* E_{bx} - E_{ay}^* E_{by} \quad (4)$$

$$A_{2,int} = E_{ax}^* E_{by} + E_{ay}^* E_{bx} \quad (5)$$

$$A_{3,int} = j(E_{ax}^* E_{by} - E_{ay}^* E_{bx}) \quad (6)$$

In practical measurements, the observed polarization state $\langle S_i \rangle$ is subject to the finite integration time T of the polarimeter. Furthermore, the time-averaged interference term is attenuated by the fields' mutual coherence, which decays exponentially as a function of the sources' linewidths $\Delta\nu$, so that broadband sources may be used to mitigate the parasitic SOP fluctuations.

The equations above were used to simulate the hybrid SMF-HCF link depicted in Fig. 1(a), where the silica-air interfaces remain a critical focal point. Despite significant efforts to minimize Fresnel reflections at these interfaces, which typically limit the performance of optical time-domain reflectometry (OTDR)^[15], marginal reflections can induce multipath interference (MPI) when narrow-linewidth light is used. In this regime, a delayed weak field (E_b) interferes with the primary field (E_a). If the SOPs are aligned ($S_a \parallel S_b$), phase fluctuations result in maximum intensity modulation with a stable SOP. Conversely, when the fields are orthogonal ($S_a \perp S_b$), intensity remains constant while SOP angle excursion is maximized. To characterize this

worst-case scenario of phase-induced SOP noise, we simulate a sinusoidal phase variation between E_a and E_b with $S_a = [1, 0, 0]$ and $S_b = [-1, 0, 0]$.

Fig. 1(b) illustrates SOP angle variations for multiple MPI levels. These SOP variations trace circular trajectories on the Poincaré sphere (Fig. 1(c)). The orientation and radius of the trajectories are fundamentally governed by the relative SOPs and the power ratio (MPI level) of the interfering fields. The peak MPI-induced SOP angular excursion, ΔSOP_{MPI} , is plotted in Fig. 1(d), which serves as a fundamental upper bound for false SOP readings induced by MPI, effectively defining the system's SOP noise floor. For example, a -40 dB MPI level – the same level reported in^[11] – may induce SOP variations up to 0.04 rad, so that fluctuations below this level can be attributed to phase-induced SOP noise. In^[11], when the fiber was exposed to weak vibrations (manual cable hitting), SOP angle variations remained under 0.02 rad (star symbol in Fig. 1(d)), making it impossible to distinguish hitting events from MPI-induced SOP fluctuations. Distinction was only possible when analyzing the rate of SOP (RSOP) changes, which would obviously enhance fast phase-changing events against the phase-noise background when ultra NLL are used (100 Hz linewidth used in^[11]).

These results demonstrate how marginal parasitic reflections translate phase instability into significant SOP wander, potentially overestimating a fiber's vibration sensitivity. However, this effect is highly coherence-dependent; as mutual coherence decreases, the SOP trajectory on the Poincaré sphere shrinks and eventually collapses toward a single point. Repeating the simulation but in the incoherent regime, $\Delta\nu = 10$ GHz (Fig. 1(e)), confirms that broadband sources effectively mitigate phase-induced SOP fluctuations.

To validate the simulations, the experiment depicted in Fig. 2(a) was conducted. An SMF-based unbalanced (30 m) Mach-Zehnder interferometer (MZI) was used to isolate phase-induced SOP changes from true polarization variations. The setup employed 100 kHz and 10 GHz linewidth sources. Different MPI levels were emulated through a variable optical attenuator (VOA). A polarization scrambler was placed in the signal arm to provide a reference for actual SOP changes, allowing a direct comparison between true polarization events and phase-induced noise.

With the scrambler off, SOP angle variations for both light sources (Fig. 2(b-c)) revealed that the narrower linewidth light exhibited a significantly higher noise level in the absence of polarization changes, confirming phase-to-polarization conversion. SOP angular excursions across varying MPI levels matched the theoretical upper bound, with the 10 GHz source results indicating an SOP

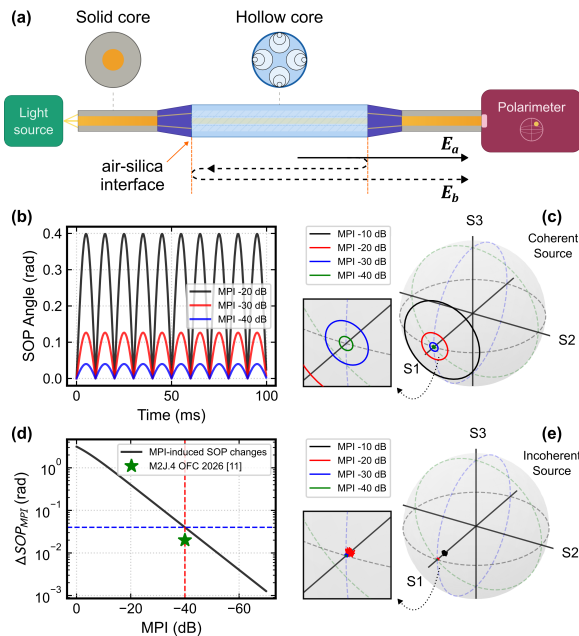


Fig. 1: (a) Simulation setup. (b) Phase-induced SOP angle variations and their circular trajectories on the Poincaré sphere (c). (d) SOP angle variations vs MPI. (e) Mitigation of SOP angle fluctuations with broadband source.

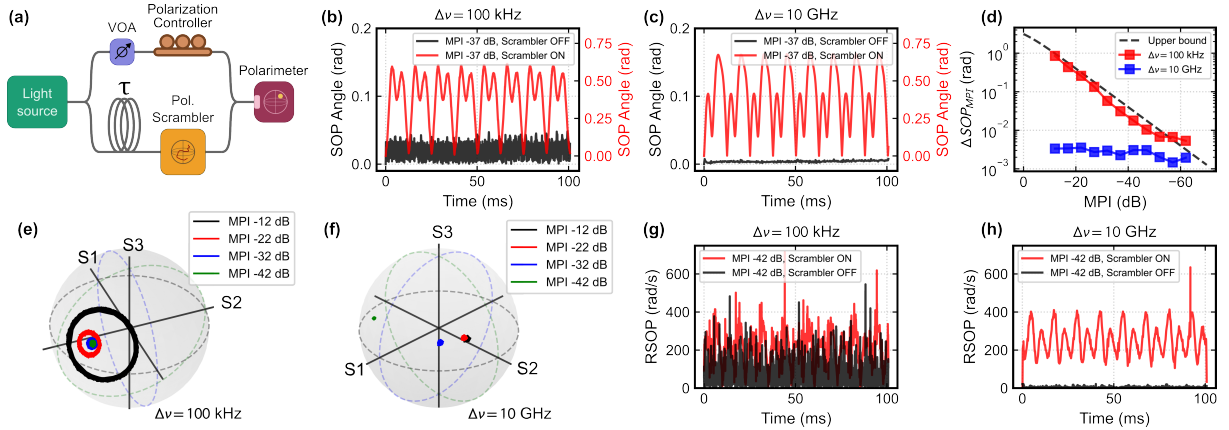


Fig. 2: (a) Experimental setup. (b–c) SOP angle fluctuations. (d) SOP excursion vs. MPI level. (e–f) Poincaré sphere trajectories for coherent and incoherent regimes. (g–h) RSOP variations.

noise floor that is decoupled from the MPI magnitude (Fig. 2(d)). On the Poincaré sphere, the narrow-linewidth source produced circular trajectories, while the broad-linewidth source remained stationary (Fig. 2(e–f)). Furthermore, the phase-induced RSOP for the narrow-linewidth case exceeded 200 rad/s (Fig. 2(g)). In contrast, the wide-linewidth source remained typically below 10 rad/s, providing noise-free RSOP traces (Fig. 2(h)).

SOP Vibration Sensitivity in NANF

For a rigorous characterization of SOP vibration sensitivity, we utilized the setup in Fig. 3(a). A polarized broadband source (12.5 GHz) was launched into 10 km of NANF, from which 0.5 m were coiled and taped on top of a carton box, having a cellphone inside emitting a 275 Hz acoustic tone at about 60 decibels, a level near the threshold of human hearing. A polarization controller (PC) adjusted the input SOP, while a polarimeter captured the response. Air-silica Fresnel reflections were estimated at -60 dB, with negligible impact due to the use of broadband source. Comparative tests were performed by replacing the NANF with an equivalent length of standard SMF.

A careful sensitivity comparison between HCF and SMF requires not only to use the same acoustic perturbation and coiled lengths, but also ensuring proper alignment between the probing light SOP at the perturbed section and the acoustically-induced birefringence axis. This aspect is depicted in Fig. 3(b), where varying the input SOP via the PC led to significant fluctuations in sensitivity to the 275 Hz tone. While Fig. 3(b) displays results for the NANF, similar behavior was observed for the SMF. Although the exact SOP at the perturbed section was not directly measured, the output SOP states (six examples shown in Fig. 3(c)) confirm a broad rotation of the input polarization. Given that the induced birefringence axis is unknown, the PC was swept through 24 distinct states for each fiber to maximize the 275 Hz response. The resulting waveforms and spectra for the maximum

sensitivity cases are shown in Fig. 3(d)–(e), where time-domain waveforms were high-pass filtered (100 Hz) and interpolated for better visualization. These results indicate that, contrary to the two-order-of-magnitude enhancement reported in^[11], NANFs exhibit an SOP vibration sensitivity comparable to that of standard solid-core fibers.

Conclusions

By utilizing an MPI-resilient characterization method that accounts for the alignment between probing SOP and acoustically-induced birefringence axis, we demonstrate that NANF SOP vibration sensitivity is comparable to that of solid-core fibers, contradicting a previous report of two-order-of-magnitude enhancement. We show that such discrepancies arise from marginal MPI (< -40 dB), which induces parasitic phase-to-SOP noise when using narrow-linewidth lasers.

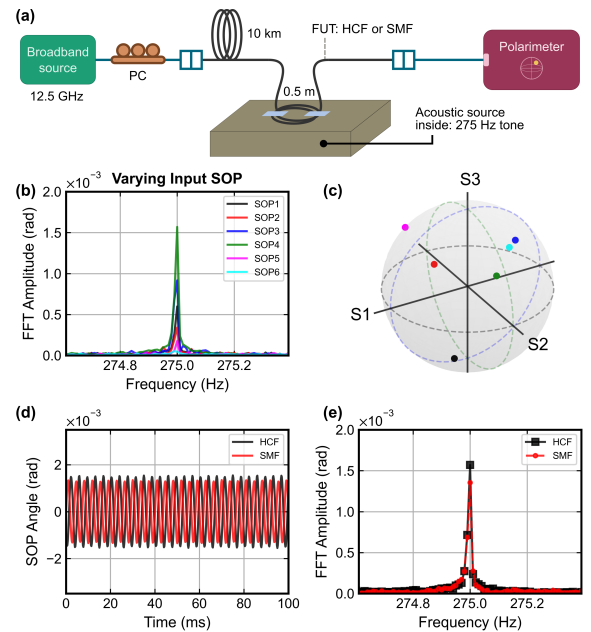


Fig. 3: (a) Setup. (b) SOP angle spectra for multiple input SOPs, and their location in the Poincaré sphere (at the detection end) (c). (d) SOP angle waveform fluctuations for HCF and SMF, and their corresponding spectra (e).

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